

Project 8 : AQI Meter

Introduction:

Pluto X2, MQ-135 sensor returns values in Analog as well. In this project we are going to look how to make use of them and use color combinations to try imitate a AQI range color values.

Challenge:

Build a program to make a AQI meter.

Add-On used:

The MQ135 is a gas sensor used to detect Ammonia (NH₃), Benzene, Alcohol, Smoke, Sulfur Dioxide (SO₂), Carbon Monoxide (CO), and general air pollutants (VOCs). It return both analog and digital values



Connection:

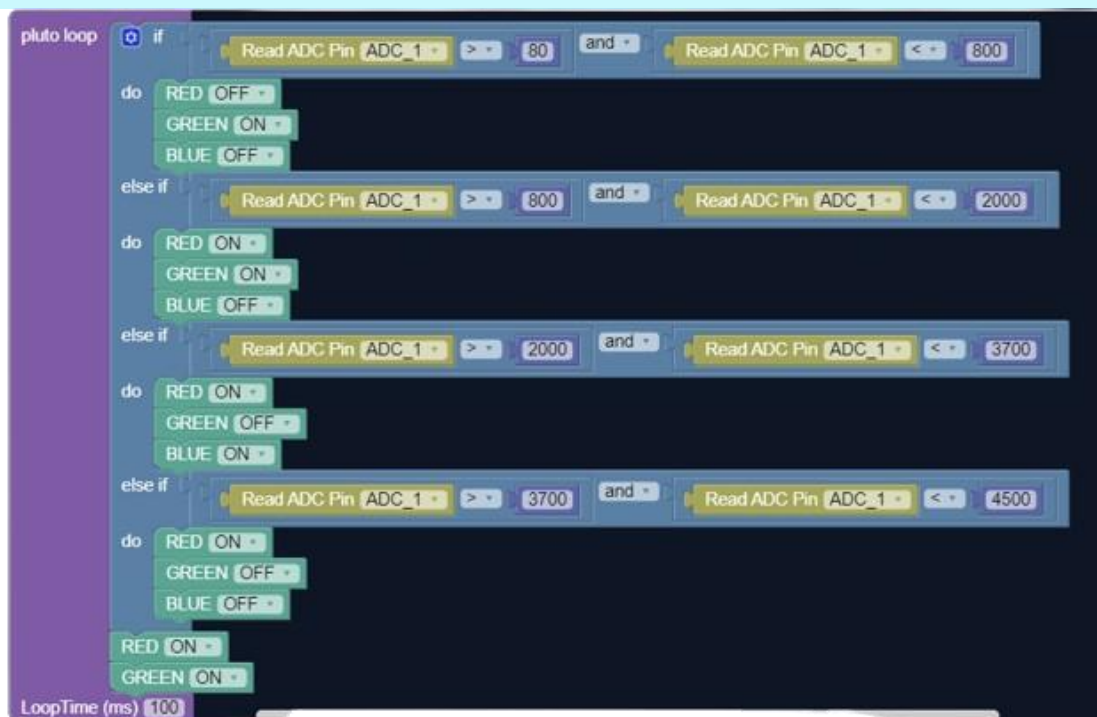
Connect the MQ-135 sensor on the ADC port of Primus V5/X2 board

Primus V5/X2 pin	Peripherals
VCC on ADC port	VCC on MQ-135
GND on ADC port	GND on MQ-135
Signal on ADC port	A0 on MQ-135

Approaching the Problem:

We divide different values coming from the MQ-135 sensor and based on them manipulate colors on indicator LED.

Solution:



Select Receiver Mode Rx_ESP

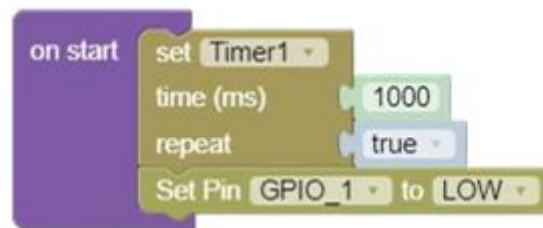
on start BLUE ON

on stop GREEN ON

Explanation of Solution:

On Start Loop:

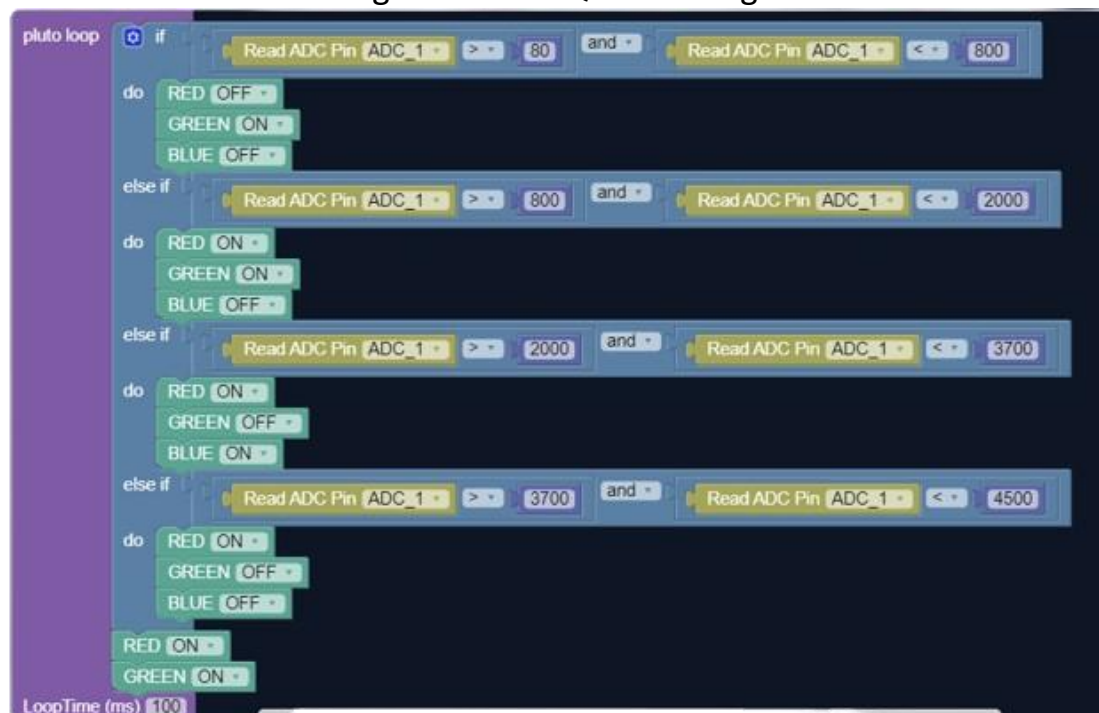
We set Blue led on



Pluto Loop:

MQ-135 returns value between 0 – 4096, but majorly returns within 500-1200 range (for smoke). So we have divided the range into 4 parts.

more the clean air, value tends toward 500 and more polluted the air, value goes towards 1200. In the loop given below, it changes the color of indicator LED showcasing different AQI color ranges.



On stop Loop: Green LED turns on.



Fig : Pluto with MQ-135 Add-on

Try further!

- Make buzzer buzz at melodies for different pollution levels.
- Try checking out levels of pollution at different heights in indoor/outdoor environments.